Ronel Herzass

ENGINEERING INTERN

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ABOUT

Third-year EE student with hands-on research experience in signal processing and embedded systems. I write C++ and Python for real hardware — from FEL simulation optimization to custom PCB design — and I bring the same rigor to teaching, having developed DSP education tools used in classroom settings.

SKILLS

Python	NumPy	Analog & Digital Circuit Design
C/C++	SciPy	Digital Signal Processing
MATLAB	Matplotlib	FFT & Spectral Analysis

EDUCATION

B.Sc. Electrical Engineering - Holon Institute of Technology | 2022 - 2028

Specialization in Signal Processing and Communications with coursework in DSP, random signals & noise, and communication systems.

PROJECTS

SonicScope – Real-Time Audio Signal Analysis Tool

GitHub | Python, NiceGUI, SciPy, NumPy, Plotly

Built an interactive audio signal analyzer featuring real-time FFT visualization, Butterworth filter implementation (high/low/band-pass/stop), and asynchronous I/O using sounddevice. Architected modular pipeline separating DSP algorithms (scipy.signal) from GUI logic, and debugged stereo-to-mono conversion affecting playback fidelity through proper NumPy array handling.

Free Electron Laser Simulation Optimization - Schlesinger Accelerators Center

Research Assistant

Ported a particle beam simulation from Python to C++ under faculty supervision, optimizing memory management and computational efficiency for electron behavior modeling in FEL research. Implemented data pipelines for processing accelerator parameters (energy, momentum, phase) and contributed to electron accelerator maintenance operations.

Automated Battery Switching System - Motor Efficiency Research Lab

Research Assistant

Designed and implemented a custom PCB interface with Arduino firmware to create an automated switching system for battery charging devices in SOC/SOH research, utilizing Arduino, PCB design tools, and C programming. Extended MATLAB analysis scripts for motor parameter characterization, contributing to battery efficiency optimization studies.

EXPERIENCE

Science Instructor - Technoda Medicine and Science Museum **2023 - Present**

Improved student comprehension of complex STEM concepts measured by enhanced engagement and project completion rates by designing hands-on experiments and mentoring FLL robotics team through parallel workflow strategies across programming, mechanical, and research sub-groups.

LANGUAGES

English - Native
Hebrew - Native

ISRAELI AIR FORCE - SPOKESPERSON UNIT 2017 – 2020

Technical Branch spokesperson in a high-pressure military environment, coordinating time-sensitive communications across multiple command levels with strict accuracy requirements.